

The Impact of Climate Policy Uncertainty on Stock Volatility: Evidence from China's eight high-carbon emission industries

Author names and affiliations.

Xia Yang^{1,2},

(1,Hubei University of Economics,Collaborative Innovation Center for Emissions Trading System Co-constructed by the Province and Ministry,Wuhan China,430205;2,Hubei University of Technology, School of Economics and Management, Wuhan China, 430068,yangxiaw@163.com)

Lincheng Jiang²

(2,Hubei University of Technology, School of Economics and Management, Wuhan China, 430068)

Zhaojie Yang^{2*},

(2,Hubei University of Technology, School of Economics and Management, Wuhan China, 430068, yangzhaojie22@gmail.com)

Corresponding author.

Zhaojie Yang^{2*},

(2,Hubei University of Technology, School of Economics and Management, Wuhan China,430068,yangzhaojie22@gmail.com)

Present address.

Hubei University of Technology, School of Economics and Management, Wuhan China,

Declaration of Interest.

none

The Impact of Climate Policy Uncertainty on Stock Volatility: Evidence from China's eight high-carbon emission industries

Abstract: In the context of global response to climate change, climate policy uncertainty induces changes in stock market volatility, but whether there are differences in its impact on volatility of different industries remains to be explored. This paper constructs a Chinese climate policy uncertainty index based on massive Chinese newspaper text data using Word2Vec algorithm, and explores the impact of the Chinese climate policy uncertainty index on the volatility of eight high-carbon emission industries in the stock market using the GARCH-MIDAS model. The results indicate that the Chinese Climate Policy Uncertainty has no significant impact on the non-ferrous metal, building material, and chemical industries, a weak impact on the paper and petrochemical industries, and a significant impact on the transportation, steel, and electric power industries. Furthermore, the results of DM test and Direction-of-Change(DoC) test indicate that the Chinese Climate Policy Uncertainty Index can effectively improve the model's forecasting accuracy of industry volatility. The results of the study help policy-making departments and investors to take targeted measures to cope with the market volatility caused by climate policy uncertainty.

Key words: Climate policy uncertainty; Industry stock volatility; GARCH-MIDAS model; mixed frequency data

1. Introduction

Climate change is one of the most serious challenges facing the globe in the 21st century and is a major concern for governments. In order to cope with the risk of climate change, governments formulate a variety of policies from time to time to guard against the impacts of climate risks, which may impact the economy and financial markets to varying degrees (Paroussos et al, 2019^[1]) and affect the volatility of financial markets and asset prices (Tian et al, 2022^[2]). As climate policy uncertainty increases, firms may need to pay higher costs and face more uncertainty to realize a low-carbon transition, thereby increasing business risks and affecting firm value.

While an increasing number of studies have examined the relationship between climate policy uncertainty and stock market performance, the majority tend to focus on the effects of such uncertainty on broad market indices or thematic indices, such as those tracking renewable energy, thereby predominantly addressing macro-market dynamics. This leaves a notable gap in the literature regarding sector-specific analyses, particularly in the context of industry heterogeneity. This gap is especially evident in the Chinese market, where pronounced policy interventions play a critical role. For instance, the technological trajectory of the power sector, a central component of the carbon market, is significantly influenced by policy directives, whereas the volatility within the building materials sector is more strongly driven by fluctuations in the real estate cycle. In actual investment, investors often construct sector portfolios based on an analysis of different sectors, and the introduction of climate policies can affect investors' risk assessment of their sector portfolios, thus changing their investment decisions. For example, the cement industry has a tough time decarbonizing due to the limitations of raw materials and production processes, a feature that may cause green investors to exit the industry altogether and instead make additional investments in electric power companies that are actively promoting the research and development of their own clean energy technologies. In addition, China's stock market is now increasingly characterized by structural market, with more frequent rotation between sectors, and the study of industry-specific indices can provide more accurate and effective information for the main body of market decision-making. Therefore, the impact of climate policy uncertainty on the volatility of different industries may differ.

China's active promotion of the 'dual-carbon' goals provides a critical framework for exploring the differential impacts of climate policy uncertainty across various industries. The 'dual-carbon' goals aim for carbon peaking by 2030 and carbon neutrality by 2060. In pursuit of these objectives, eight major high-carbon-emitting industries—namely, nonferrous metals, paper, electricity, aviation, building materials, petrochemicals, iron and steel, and chemicals—accounting for 75% of China's total carbon emissions, will gradually be incorporated into the carbon emissions trading system during the 14th Five-Year Plan period. As these industries are integrated into the carbon market, the climate policy environment they face will undergo significant shifts, leading to varying impacts across sectors. Consequently, the volatility of stocks

in these industries is likely to be influenced to differing extents, sparking considerable attention and research within the academic and policy communities at the same time. With continued high levels of concern, climate policy uncertainty affects the investment strategies of investors holding assets related to the eight sectors, and the corresponding buying and selling behaviors cause changes in short-term stock prices, which in turn cause volatility in the sector indices (Chen et al, 2023^[3]). At the same time, the impact of climate policy uncertainty tends to exist for a longer period of time, which is because investors will revalue companies facing more uncertainty, and this revaluation process is not only accompanied by the consideration of the profitability, growth, investment planning and other aspects of the enterprise, but also includes the investor's own judgment on the industry development trend, the next step of the policy expectations and other aspects. Therefore, in order to study the impact of climate policy uncertainty on the stock volatility of eight industries without losing the richer high-frequency market volatility information, this paper adopts the GARCH-MIDAS model, which has good stock volatility prediction ability, to simultaneously incorporate low-frequency monthly macro information and high-frequency daily volatility data to analyze, and to examine the predictive effect of Chinese Climate Policy Uncertainty (CCPU) Index on the stock volatility of the industry.

This paper makes two primary contributions. First, in terms of index construction, it employs the Word2Vec algorithm to derive a comprehensive set of keywords for the development of the China Climate Policy Uncertainty Index. This approach addresses the limitations of subjective keyword selection biases present in prior studies, while also ensuring that the index better reflects China's climate policy uncertainty by incorporating official policy documents and mainstream media sources. This refined index provides a more accurate and holistic representation of climate policy fluctuations, enabling investors to assess market risks and prospects more effectively and adjust their investment strategies accordingly. Second, from an industry-specific perspective, the study identifies significant heterogeneity in the impact of China's climate policy uncertainty on stock volatility across different sectors. This finding deepens the understanding of the relationship between climate policy uncertainty and stock market dynamics and offers valuable insights for both government agencies and investors in recognizing the potential risks and opportunities within various industries.

The subsequent structure of this paper is organized as follows: the second part provides a literature review; the third part introduces the methodology for constructing the Chinese Climate Policy Uncertainty Index and describes the empirical modeling and testing methodology; the fourth part discusses the results

of the empirical study and conducts a robustness test; and finally, the main conclusions and recommendations are presented.

2. Literature Review and Theoretical Analysis

2.1 Literature Review

Scholars have long been concerned with policy uncertainty, with early studies such as Friedman(1968^[4]), Rodrik(1991^[5]), and Hassett and Metcalf(1999)^[6] identifying the adverse effects of monetary, fiscal, and regulatory policy uncertainty on the economy. In recent years, Pástor and Veronesi(2013)^[7] modeled the theoretical link between volatility, policy uncertainty, and stock market volatility, and Born and Pfeifer(2014)^[8] examined the adverse effects of policy uncertainty in a DSGE model. Hassan et al(2019)^[9] found that higher policy uncertainty slows the growth of employment growth and increases stock price volatility in U.S. firms.

In the context of addressing climate change, in order to avoid climate change risks and promote green and sustainable economic growth, the government has introduced and adjusted a series of climate policies (Wesseh et al, 2022)^[10], thus elevating climate policy uncertainty (Borenstein et al, 2019^[11]). Climate policy uncertainty can cause larger and longer-term impacts on enterprises, on the one hand, climate policy uncertainty can exacerbate the imbalance between supply and demand fundamentals in the commodity market (Liu et al, 2023^[12], Ding et al, 2023^[13]), making it difficult for relevant business entities to form accurate expectations for their production and R&D, etc. plans, and affecting enterprise cash flow and performance; on the other hand, since enterprise investment is irreversible, under the climate policy uncertainty, high carbon emitting enterprises may adopt a "wait-and-see" strategy and reduce their R&D investment (Hoang, 2022^[14]) and external financing (Ren et al, 2022^[15]). The impact of climate policy uncertainty on firms can exacerbate stock market volatility and is an important channel through which transition risk affects stock markets (Kunreuther et al, 2013^[16]).

In recent years, there has been a rapid increase in research on the relationship between climate policy uncertainty and financial markets. From the aspect of research perspectives, at the macro level, in the absence of market expectations, irregular releases of climate policies such as low-carbon transition policies, green finance, and green energy subsidies may have a complex impact on stock prices in different sectors, and many assets will be revalued in the short term, which will be susceptible to cyclical losses (Batten, 2018^[17]). Ren et al (2023)^[18] show that whenever there is an extreme climate event or a major climate policy change, the causality between climate policy uncertainty and the related market such as green significantly rises. For stock market investors, climate policy uncertainty is an important factor influencing

their investment decisions. Investors' concern about climate policy uncertainty increases their sensitivity to unknown risks, which leads to irrational behavior (Guo et al, 2022^[19]) and increases the downside risk of stocks (Chen et al, 2023^[20]). Xu et al(2022)^[21] find that climate policy uncertainty predicts renewable energy market returns well, and the higher the index the better the predictive performance. The findings of He and Zhang (2022)^[22] suggest that climate policy uncertainty is a strong predictor. And at the industry-specific level, Yan and Cheung (2023)^[23] examine the dynamic effects of climate policy uncertainty and coal price on carbon price in China. A study by Coeure (2018)^[24] found that carbon emission policies of sudden tightening could lead to a repricing of carbon-intensive assets, which could hit fossil fuel prices and the value of related firms, which in turn affects their solvency and the financial position of investors holding their debt or equity. Since different industries are not affected by climate policy uncertainty to the same extent, investors may actively adjust their investment portfolios to avoid risks or chase profits, and the large flow of funds between industries can produce changes in volatility and affect market stability. It can be seen most studies on the relationship between climate policy uncertainty and stock market volatility have examined its impact on the volatility of stock market composite indices or thematic indices such as energy (Xu et al, 2023^[25], Liang et al, 2022^[26], Chen et al, 2023^[27]), and empirical studies on the impact on specific industries are still limited. Since different industries are not affected by climate policy uncertainty to the same extent, investors may actively adjust their investment portfolios to avoid risks or chase profits, and the large flow of funds between industries can produce changes in volatility and affect market stability. In studies of stock markets, it is important to distinguish between different industries (Baumöhl and Lyócsa, 2017^[28], Lu et al, 2022^[29]), so there is a need to analyze the effects of climate policy uncertainty, a macro-variable, on the volatility of different industries in the stock market as well as the potential mechanisms behind these effects.

In terms of the construction methodology, existing studies in measuring climate policy uncertainty have constructed indices based on manually set sets of keywords (Yan and Cheung, 2023^[23], Xu et al, 2023^[25]), whose accuracy relies on the setup rules and manual auditing, leading to some subjective selection bias in the results. Gavrilidis(2021) ^[35] developed a Climate Policy Uncertainty (CPU) index, derived from the frequency of climate policy-related keywords in U.S. newspapers. This index has become a widely utilized tool in studies examining the impacts of climate policy uncertainty on macroeconomic conditions, energy markets, financial stability, business operations, and investment strategies. However, the U.S. CPU index is based on the selection of U.S. newspapers, and the climate policy keywords it incorporates may not be fully applicable to China's unique context, thus limiting its ability to accurately

capture the dynamics of climate policy uncertainty in China.

In general, the literature on climate policy uncertainty has been rapidly increasing in recent years, and the impact of climate policy uncertainty on the economy and finance has been explored from various perspectives, such as energy market, corporate decision-making, and asset pricing, etc. However, the measurement method of the climate policy uncertainty index is still relatively single and relies on subjective judgments, and the research on the specific impacts of climate policy uncertainty on different industries is also very limited. Therefore, this paper constructs an index of Chinese climate policy uncertainty based on deep learning of keywords from a large number of newspaper texts, and studies the relationship between Chinese climate policy uncertainty and the volatility of stocks in eight high-carbon emitting industries as samples.

2.2 Theoretical Analysis

Against the backdrop of the global emphasis on climate change, climate policies have been introduced more frequently and covered a wider range of areas, and climate policy uncertainty is also increasing. The implementation and adjustment of climate policies will not only change the supply and demand situation and investment and financing activities of the affected industries, but also be known by all sectors of the society due to the public nature of the policies, resulting in climate policy uncertainty affecting stock volatility mainly through the channels of industry operating fundamentals and investor behavior.

Climate policy uncertainty will change the supply and demand situation of the affected industries. On the one hand, restrictive climate policies, such as total volume control and dual control of energy consumption, will inhibit or increase the supply of upstream raw materials or the production of downstream products in these industries; on the other hand, changes in social and consumer preferences brought about by changes in climate policy will affect the demand for related products. Changes in supply and demand will directly impact the price of related products through the substitution effect and aggregate demand effect, affecting the business performance of enterprises and even the whole industry. At the same time, climate policy uncertainty can affect the investment and financing behavior of enterprises in these industries through the anticipation effect. The existence of climate policy uncertainty will make it difficult for enterprises to form accurate expectations of the consequences of climate policy, and the financial friction effect will make it more difficult for enterprises to raise funds, enterprises often choose to delay investment, R&D investment and financing needs will be reduced accordingly (Blyth et al, 2007^[30]). As a result, the impact of climate policy uncertainty can have an important effect on the cash flows of the affected industries, and equity investors will continuously adjust their expected returns and change their

investment strategies based on their performance, both in terms of aggregate portfolio changes and structural adjustments, which will cause capital flows in the stock market and increase volatility in the industry segments. Further, superimposed on the spillover effects of costs, profits and investments upstream and downstream of these industries that are also affected, such volatility can also be transmitted across markets through trade networks and linkages between financial markets, thus exacerbating the degree and duration of volatility (Benhabib et al, 2016^[31]).

The promulgation of climate policies is often communicated through various channels of information dissemination and triggers investor attention, either directly or indirectly. The direct route is when investors directly notice the policy content and change their investment behavior accordingly, while the indirect route is when the policy promulgation leads to changes in market opinion and expectations, which trigger investors' attention. In terms of the direct route, investors will directly notice the promulgation of climate policies mostly because the industry they invest in is closely related to policy changes, for example, investors in the photovoltaic industry will always pay attention to the promulgation of policies such as the management of power stations, innovation support, and so on. The introduction of policies will cause investors to reassess the relevant industries and adjust their investment portfolios, which will lead to the flow of funds among different industries in the stock market, resulting in fluctuations in industry sectors. In terms of indirect pathways, the limited attention theory suggests that investors' attention span is limited (Hirshleifer and Hong Teoh, 2003^[32]) and that investors are more likely to pay attention to recent news that follows the trend of public opinion, is frequent, or has been widely discussed, and that if there is increasing coverage and discussion of climate policy change, then investors across the market will be attentive to this information. According to the finite rationality hypothesis of behavioral finance theory, investors cannot always maintain rational and correct thinking when making decisions, and investors in the climate policy uncertainty environment will increase their sensitivity to the related risks, and therefore will show more finite rationality characteristics such as overconfidence and the herd effect (Guo et al, 2022^[19]), which will result in a large number of trading behaviors such as diversification and follow-through investment. behaviors that exacerbate the volatility of industry stocks. In addition, changes in stock volatility itself will also affect investors' attention and behavior, and volatility will further increase under the mutual influence of the two. In recent years, there has been a significant increase in the number and frequency of climate policies, which has attracted a great deal of investor attention, and therefore exacerbated the market volatility of the relevant industries.

3. Climate Policy Uncertainty Index Construction and Modeling

Introduction

3.1 Climate Policy Uncertainty Index Construction

Existing studies on policy uncertainty mostly use the measurement method proposed by Baker et al(2016)^[33], on the basis of which Huang et al (2020)^[34] and Gavrilidis (2021)^[35] constructed the Chinese economic policy uncertainty index and the U.S. climate policy uncertainty index, respectively. In order to combine China's actual national conditions, this paper further constructs Chinese climate policy uncertainty index based on domestic newspaper data. All the articles from January 1, 2008 to May 31, 2023 were obtained.

Second, three terminology sets were constructed for "climate", "policy" and "uncertainty". The terms in the "uncertainty" term set are similar to the uncertainty terms used in most of the existing literature on policy uncertainty measurement. The acquisition of the "climate" and "policy" term sets is different from most of the existing studies that use human settings and manual auditing to obtain keywords, this paper applies the Word2Vec algorithm proposed by Mikolov et al(2013)^[36] to identify terms related to climate and policy. Word2Vec algorithm is an unsupervised deep learning method that does not require manual labeling of data and can autonomously learn the intrinsic representation of the data, so the collection of terms that are highly related to the two words climate and policy can be obtained by mapping each word to a high-dimensional space through the training of the model, and by measuring the inter-word correlation based on the vectorial distances of different words. In this paper, we choose the top-level documents released by China, such as the National Program for Addressing Climate Change, the White Paper on Policies and Actions for Addressing Climate Change, and the Report on the Work of the Government, as the corpus of the model. In the second step, the preprocessing results are converted to one-hot coded representations to obtain the word vector of each word, and then the words are mapped to a 50-dimensional word vector space by the Skip-gram architecture of Word2Vec for training. In the third step, after completing the training, the cosine similarity between the words is calculated to find the words that are closest to "climate" and "policy" to form a term set. All three term sets are shown in Table 1.

Table 1. The term set of Chinese climate policy uncertainty index

Category	Keyword
Climate	Carbon emissions / Carbon dioxide / Climate change / Climate risk / Global
	warming / Greenhouse gases / Green energy/
	Environmental protection/ Carbon Neutrality/ Carbon Peaking

	State Council/ Central/ People's Congress/ Securities and Exchange Commission/ Ministry of Ecology and Environment/ Ministry of Environmental Protection/ National Development and Reform Commission/ National Energy Administration/ Policy/ Statute/ Law/ Strategy/ Regulation/ Institution/ Measure/ Supervision/ Rectification/ Reform/ Prevention
Policy	
	Uncertain/Uncertainty/Not certain/Unspecified/Unknown/Hard to predict/Hard to tell/Unclear/Unsure/Difficult to anticipate/ Difficult to estimate/ Difficult to predict/ Uncertainty
	Difficult to determine/ Unpredictable/Unable to estimate/unforeseeable

Finally, the number of articles per month that contained at least one term from each of the three term sets simultaneously was calculated in each newspaper using python, and articles were manually screened to avoid the presence of misreported articles. Next, the frequency of articles was standardized and normalized with reference to the method of Baker et al. (2016)^[33], and the formula was calculated as follows:

$$Y_i = \frac{1}{N} \sum_{n=1}^N \frac{X_{in}}{\sigma_n} \quad (2)$$

$$CCPU = \frac{IY_i}{\sum_{i=1}^I Y_i} \times 100 \quad (3)$$

where X_{in} is the frequency of the n_{th} eligible article in the i_{th} month, σ_n is the standard deviation of the series, and CCPU is the monthly Chinese climate policy uncertainty index.

Figure 1 depicts the trend of Chinese climate policy uncertainty index from 2008 to 2023. Before China formally proposed the "dual-carbon" goal, the index peaked significantly after the release of documents related to climate change in the 12th Five-Year Plan and the 13th Five-Year Plan. Before China formally proposed the "dual-carbon" goal, the index peaked after the release of documents related to climate change in the 12th Five-Year Plan and the 13th Five-Year Plan, and after the introduction of the "dual-carbon" goal in 2020, the attention paid by major newspapers to climate issues increased significantly, and the index showed an upward trend.

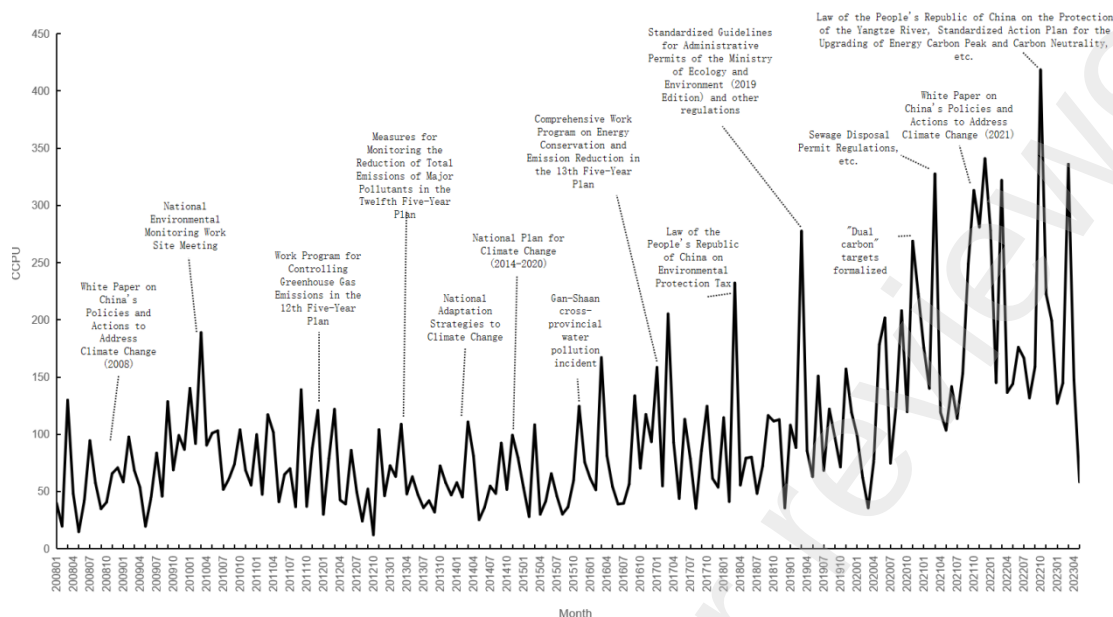


Figure 1. Trends in Chinese Climate Policy Uncertainty Index and Matching Realistic Major Events

This paper compares the constructed index of Chinese climate policy uncertainty with two uncertainty-related indicators. The climate policy uncertainty index constructed by Gavrilidis (2021)^[35] based on U.S. newspapers has received widespread attention and recognition. Figure 2 shows that the correlation coefficient between the index constructed in this paper and it is 0.57. With the increase in global attention to climate change, the Both of them show a rising trend. Figure 3 shows the relationship between this paper's index and the realized volatility of China's stock market SSE index, which is not correlated.

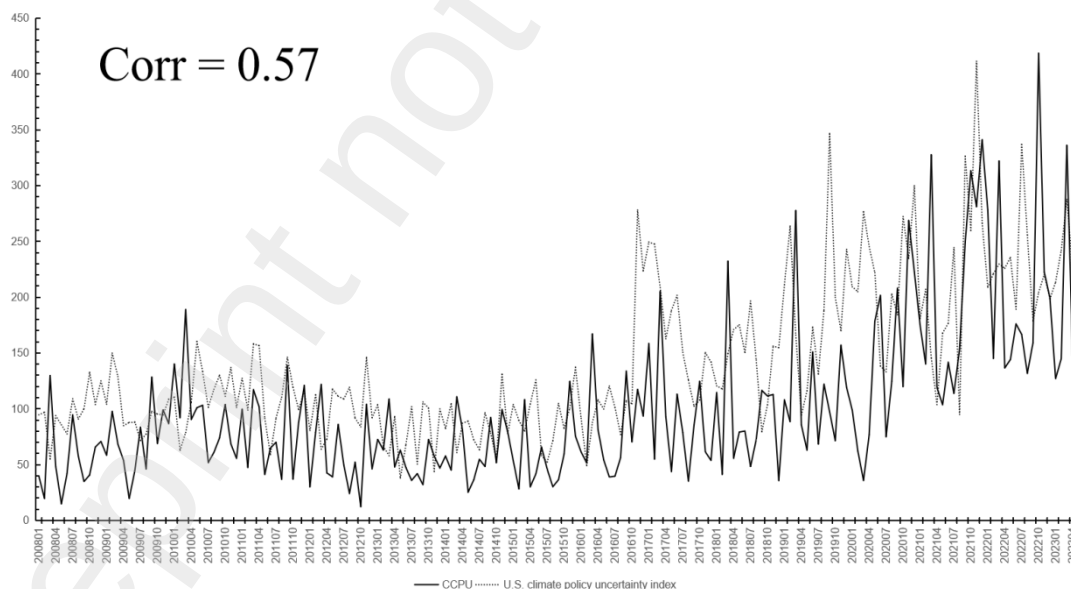


Figure 3. Climate Policy Uncertainty Index Based on Chinese Newspapers and U.S. Newspapers

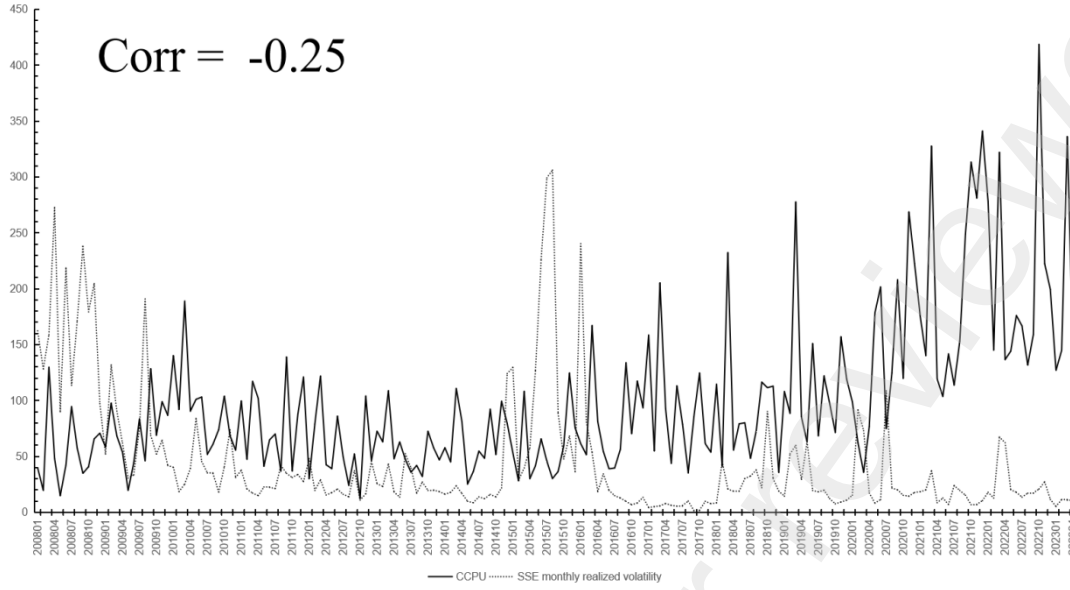


Figure 3. Chinese Climate Policy Uncertainty Index and SSE Volatility

3.2 GARCH-MIDAS model

In this paper, the Chinese climate policy uncertainty index is monthly data, and the yield series is daily data. In order to avoid the loss of effective information in high-frequency data, Engle et al. (2013)^[37] proposed a GARCH-MIDAS model based on MIDAS (Mixed Frequency Data Sampling), which decomposes the conditional variance into long and short-term components to solve the problem of data frequency inconsistency in the following mathematical form:

$$\begin{cases} r_{it} - E_{i-1,t}(r_{it}) = \sqrt{\tau_t g_{it}} \varepsilon_{it}, \forall i = 1, 2, \dots, N_t \\ \varepsilon_{it} | \psi_{i-1,t} \sim N(0, 1) \end{cases} \quad (4)$$

where r_{it} is the return on day i of month t ; $E_{i-1,t}(r_{it})$ is the conditional mean at a given moment; $\varepsilon_{it} | \psi_{i-1,t} \sim N(0, 1)$ is the set of historical information up to day $i-1$ of month t , τ_t denotes the long-term component, and g_{it} denotes the daily short-term component.

Let $E_{i-1,t}(r_{it}) = \mu$, then equation (3) can be expressed as:

$$r_{it} = \sqrt{\tau_t g_{it}} \varepsilon_{it} + \mu, \forall i = 1, 2, \dots, N_t \quad (5)$$

In Eq. (4), τ_t uses the rolling window approach (which removes the restriction that τ_t is fixed in month t and allows τ to vary at a frequency of days for long-term updates), and g_{it} obeys a standard

GARCH(1,1) process inscribed by a MIDAS regression based on the monthly realized volatility

$$RV_t = \sum_{i=1}^{N_t} r_{it}^2, \text{ i.e:}$$

$$g_{it} = (1 - \alpha - \beta) + \alpha \frac{(r_{it} - \mu)^2}{\tau_t} + \beta g_{i-1,t} \quad (6)$$

$$\tau_t = m + \theta \sum_{k=1}^K \varphi_K(\omega_1, \omega_2) RV_{t-k} \quad (7)$$

where $\alpha > 0$, $\beta > 0$, $\alpha + \beta < 1$, and K are the maximum lag orders of the low-frequency variables, and $\varphi_K(\omega)$ is a weighted kernel function based on the Beta function, for which the beta polynomials are more flexible and adaptable to a variety of lag structures, with the expression:

$$\varphi_k(w) = \frac{\left(1 - \frac{k}{K}\right)^{w-1}}{\sum_{j=1}^K \left(1 - \frac{j}{K}\right)^{w-1}}, \forall k = 1, \dots, K \quad (8)$$

In order to examine the impact of Chinese climate policy uncertainty on the eight industries, this paper simultaneously uses the Chinese Climate Policy Uncertainty Index to replace the realized volatility RV, thus obtaining the GARCH-MIDAS-CCPU model:

$$\tau_t = m + \theta_C \sum_{k=1}^{K_C} \varphi_K(\omega_{1,C}, \omega_{2,C}) CCPU_{t-k} \quad (9)$$

Equations (3) to (8) constitute a one-factor GARCH-MIDAS model, and this paper further introduces the Chinese Climate Policy Uncertainty Index (CPUI) as an exogenous low-frequency variable on the basis of Equation (6) and Equation (8), and constructs a two-factor GARCH-MIDAS model in order to capture the effects of the different factors on the long-run components of the industry volatility at the same time:

$$\tau_t = m + \theta_R \sum_{k=1}^K \varphi_K(\omega_1, \omega_2) RV_{t-k} + \theta_C \sum_{k=1}^{K_C} \varphi_K(\omega_{1,C}, \omega_{2,C}) CCPU_{t-k} \quad (10)$$

3.3 Evaluation methodology of the model

3.3.1 Diebold-Mariano (DM) test

In this paper, the DM test proposed by Diebold and Mariano (1995)^[38] is used to compare the forecasting accuracy between the models. A GARCH-MIDAS model based on the RV of realized volatility is used as the baseline model so that it can be compared with a one-factor GARCH-MIDAS-CCPU model and a two-factor GARCH-MIDAS model. The mathematical form is as follows:

$$DM = \frac{\bar{d}}{\sqrt{\text{var}(d_t)}} \sim N(0,1), d_t = e_{A,t}^2 - e_{B,t}^2 \quad (11)$$

where $e_{A,t}$ and $e_{B,t}$ are the prediction errors comparing models A and B, and $\bar{d}$ is the expectation of the sequence d_t .

3.3.2 Direction-of-Change (DoC) test

Degiannakis and Filis (2017)^[39] use the DoC test as a forecasting accuracy criterion, where DoC measures the proportion of the model that correctly predicts the direction of volatility change (higher or lower). Assuming that P_t is a dummy variable assigned a value of 1 if the direction of volatility change is correctly predicted on day t, and 0 otherwise, the equation is:

$$P_t = \begin{cases} 1, & \text{if } \sigma_t^2 > \sigma_{t-1}^2 \text{ and } \hat{\sigma}_t^2 > \sigma_{t-1}^2, \\ 1, & \text{if } \sigma_t^2 < \sigma_{t-1}^2 \text{ and } \hat{\sigma}_t^2 < \sigma_{t-1}^2, \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

Let DoC denote the proportion of correctly predicting the direction of change in volatility, i.e:

$$DoC = q^{-1} \sum_{t=m+1}^{m+q} P_t \quad (13)$$

where m denotes the in-sample estimation length and q denotes the out-of-sample prediction interval length. To investigate the statistical significance of the prediction accuracy, this paper uses the nonparametric test proposed by Pesaran and Timmermann (2009)^[40] to test for significance. The original hypothesis of this method is that the DoC value of the prediction model is less than or equal to the DoC value of the random walk, so the p-value of its P_t -statistic can be used to test whether the original hypothesis is rejected.

4. Empirical analysis

4.1 Data description and descriptive statistics

The Shenwan Industry Index is an industry stock index researched and released by Shenwan Hongyuan Securities, which is one of the strongest securities companies with comprehensive strength in China, and the industry indexes released by it are widely recognized by the industry, and many domestic securities companies carry out industry research based on the classification of Shenwan Hongyuan. Therefore, in this paper, non-ferrous metal (801050.SWI), transportation (801170.SWI), building material (801710.SWI), petroleum and petrochemical (801960.SWI), steel (801040.SWI), and basic chemical

industry (801030.SWI) in the Shenwan primary Industry Index are selected to represent the six industries of non-ferrous metal, aviation, building material, petrochemical, steel, and chemical among the eight industries, and paper (801143.SWI) and electric power (801161.SWI) in the Shenwan secondary industry index are selected to represent the paper and electric power industries. Each industry is characterized by 3,747 daily return data, and the data source is Choice database. The sample period is from January 1, 2008 to May 31, 2023. Two large bull and bear transitions in 2008 and 2015 are included in this study period, thus better reflecting the volatility characteristics at different stages.

The descriptive statistics of Chinese climate uncertainty index and the eight industry indices are shown in Table 2, where NFM, PA, EP, TR, BM, PP, ST, and BC are the indices of non-ferrous metal, paper, electric power, transportation, building material, petroleum and petrochemical, steel, and basic chemical, respectively. In terms of skewness and kurtosis, Chinese climate policy uncertainty index is right-skewed, and the eight industry indices are left-skewed, which are all characterized by "sharp peaks and thick tails". The results of the ADF test indicate that the variables are smooth time series; the results of the J-B test indicate that the nine variables do not obey normal distribution; the results of the Q(n) test indicate that the variables have obvious autocorrelation characteristics at lags 5 and 22. The results of ADF test indicate that all variables are smooth time series; J-B test indicates that all nine variables do not obey normal distribution; Q(n) results indicate that all variables have obvious autocorrelation characteristics when the lag period is 5 and 22. Therefore, the GARCH-MIDAS model can be directly used to model and analyze the stock volatility of the eight industries from the long and short-term components.

Table 2. Descriptive statistics for variables

	CCPU	NFM	PA	EP	TR
mean	100	0.014	0.012	0.008	-0.006
median	79.596	0.082	0.108	0.051	0.051
max	418.569	9.675	9.809	9.183	9.961
min	12.209	-9.633	-9.501	-8.469	-9.425
sd	70.841	2.248	1.973	1.586	1.706
skewness	1.807	-0.282	-0.555	-0.47	-0.498
kurtosis	6.669	5.243	6.285	7.706	7.192

ADF	-2.572***	-11.511***	-11.766***	-11.97***	-11.757***
J-B	204.5***	835.4***	1877***	3595***	2898.8***
Q(5)	193.0823***	25.8714***	16.7604***	19.1217***	22.1145***
Q(22)	563.379***	50.0594***	45.0943***	63.3045***	57.5537***
	BM	PP	ST	BC	
mean	0.024	-0.01	-0.01	0.012	
median	0.087	0.047	0.066	0.119	
max	9.451	9.842	9.903	8.677	
min	-9.533	-8.64	-9.786	-8.541	
sd	2.013	1.734	1.971	1.783	
skewness	-0.355	-0.265	-0.282	-0.554	
kurtosis	5.703	6.786	5.99	5.867	
ADF	-11.818***	-12.111***	-11.063***	-11.815***	
J-B	1219***	2281.4***	1444.9***	1475.2***	
Q(5)	24.103***	16.5095***	10.9522*	25.311***	
Q(22)	54.0008***	49.6905***	39.6221***	55.5503***	

Notes: *** denotes rejection of the original hypothesis at the 1% level, * denotes rejection of the original hypothesis at the 10% level, J-B test is a normal distribution test, and Q(n) denotes the lagged n_{th} order Ljung-Box Q statistic.

4.2 In-sample estimation results

4.2.1 Single factor GARCH-MIDAS model estimation

In order to study the role of shocks of each exogenous variable, the models are first estimated with full samples and the RV-based model is used as the reference model. According to Bai et al (2022)^[41], if the data frequency of the GARCH-MIDAS model is monthly and daily data, the estimated value of the lag parameter K is reasonable within 36, and in view of the existence of a longer-term impact of climate policy uncertainty on stocks, this paper selects the lag parameter K = 24, which means that the impact of each

exogenous variable on the industry stock volatility is 24 months, and the estimation result are shown in Table 3.

From the short-term part, the mean parameters of all the models are insignificant, the α and β parameters are significant at the 1% level, and the value of $\alpha+\beta$ is not greater than 1, which indicates that the returns of the eight industries have a volatility aggregation effect on short-term fluctuations, and therefore, the model constructed in this paper is stable.

From the long-term part, the parameter ω is greater than 1, which indicates that the weight coefficient of the lagged term decreases with time, i.e., the closer to the current period, the greater the influence of low-frequency variables on the index volatility. Parameter θ represents the effect of RV and chinese climate policy uncertainty index on the volatility of stocks in eight industries, which is the parameter of focus in this paper. Table 3 presents the estimation results of the eight industries based on RV and chinese climate policy uncertainty index respectively. The parameter θ of the RV-based model is significantly positive at the 1% level, indicating that the realized volatility has a large impact on the long-term volatility of the indexes of the eight industries, and exacerbates the volatility of the indexes, increases the investment risk, and is not conducive to the long-term stability of the market. The parameter θ of the model based on Chinese climate policy uncertainty index shows greater industry heterogeneity, with non-ferrous metal, building material, and chemical industry having insignificant parameter θ ; paper industry having a significant positive parameter θ at the 10% level; transportation, petrochemical, and steel industry having significant positive parameter θ at the 5% level; and electric power industry having a positive parameter θ at the 1% level, which suggests that the increase of chinese climate policy uncertainty significantly exacerbates the fluctuation of the indices of these industries.

Table 3. Estimation results of single factor GARCH-MIDAS model

	NFM		PA		EP	
	RV	CCPU	RV	CCPU	RV	CCPU
μ	0.0088 (0.032)	0.0051 (0.0318)	0.0326 (0.0269)	0.0289 (0.0269)	0.0147 (0.0196)	0.0084 (0.0191)
α	0.0721*** (0.0087)	0.0644*** (0.0065)	0.0767*** (0.0082)	0.0701*** (0.0064)	0.1025*** (0.0073)	0.0726*** (0.0045)
β	0.8788*** (0.0238)	0.9135*** (0.0085)	0.8765*** (0.0196)	0.9071*** (0.0086)	0.8292*** (0.0201)	0.9200*** (0.004)
θ	0.0197*** (0.0053)	0.0066 (0.004)	0.0178*** (0.0048)	0.0039* (0.0023)	0.0340*** (0.0035)	0.0116*** (0.0036)
ω	8.8476 (5.5536)	38.392 (50.022)	7.6468 (4.7187)	49.834 (72.779)	12.917*** (3.0122)	41.602** (16.456)

m	1.995*** (0.4175)	3.0947*** (0.4958)	1.6556*** (0.3035)	2.5149*** (0.3361)	0.4499*** (0.0839)	0.9987*** (0.364)
TR		BM		PP		
	RV	CCPU	RV	CCPU	RV	CCPU
μ	0.0023 (0.0221)	-0.0027 (0.0218)	0.0406 (0.0281)	0.0375 (0.0277)	0.0174 (0.0241)	0.0091 (0.024)
α	0.0710*** (0.006)	0.0618*** (0.0044)	0.0610*** (0.0076)	0.0519*** (0.0047)	0.0900*** (0.0082)	0.0652*** (0.0052)
β	0.8863*** (0.0174)	0.9245*** (0.0045)	0.8812*** (0.0345)	0.9356*** (0.0054)	0.8354*** (0.0249)	0.9192*** (0.0056)
θ	0.0248*** (0.0042)	0.0026** (0.0037)	0.0269*** (0.0054)	-0.0030 (0.0053)	0.0286*** (0.0036)	0.0068** (0.0034)
ω	8.7853** (3.605)	2.0015 (1.011)	14.509** (6.7808)	3.2194 (14.689)	12.146*** (4.1894)	23.529 (20.459)
m	0.8331*** (0.1659)	2.9621*** (0.4742)	1.2062*** (0.334)	3.5425*** (0.6516)	0.854*** (0.1471)	1.776*** (0.3984)
ST				BC		
	RV	CCPU	RV	CCPU		
μ	0.0063 (0.0272)	-0.0014 (0.028)	0.0285 (0.0268)	0.0159 (0.0263)		
α	0.0859*** (0.0081)	0.0763*** (0.0064)	0.0913*** (0.0087)	0.0628*** (0.0051)		
β	0.8560*** (0.0206)	0.9048*** (0.007)	0.7974*** (0.0315)	0.9194*** (0.0059)		
θ	0.0237*** (0.0043)	0.0078** (0.0038)	0.0286*** (0.003)	-0.0050 (0.0044)		
ω	9.8623** (3.8519)	39.993 (45.257)	18.219*** (4.0688)	2.0012 (2.185)		
m	1.2968***	2.2339***	0.874***	2.9919***		

(0.2468)

(0.4764)

(0.1314)

(0.5106)

Notes: Robust standard errors are in parentheses; *, **, and *** indicate rejection of the original hypothesis at the 10%, 5%, and 1% levels, respectively.

4.2.2 Double factor GARCH-MIDAS model estimation

In order to investigate the joint effect of RV and Chinese Climate Policy Uncertainty Index on the long-run components of stock volatility in eight industries, this paper extends the single factor model into a double factor model by including these two low-frequency variables at the same time. The estimation results are shown in Table 4.

The results show that the parameters of each model θ_R are significantly positive at the 1% level, again indicating that low-frequency realized volatility can significantly increase the long-term component of high-frequency volatility in the eight industry indices, exacerbating market volatility. Parameter θ_C for nonferrous metals, building material, and chemical industry is still insignificant, that for transportation industry is significant at the 5% level, and that for steel and electric power industry is significant at the 1% level, which is basically consistent with the results of the single factor model estimation. Parameter θ_C is no longer statistically significant for paper and petrochemical, suggesting that the impact of Chinese climate policy uncertainty on these two industries is relatively small.

Table 4. Estimation results of double factor GARCH-MIDAS model

	NFM	PA	EP	TR
μ	0.0090 (0.032)	0.0325 (0.0271)	0.0136 (0.0196)	0.0023 (0.022)
α	0.0708*** (0.0089)	0.0756*** (0.0081)	0.0968*** (0.0073)	0.0715*** (0.0062)
β	0.8827*** (0.0245)	0.8803*** (0.0191)	0.8457*** (0.019)	0.8837*** (0.0199)
θ_R	0.0186*** (0.0058)	0.0170*** (0.005)	0.0328*** (0.0038)	0.0238*** (0.0046)
θ_C	0.0045 (0.003)	0.0022 (0.0018)	0.0033*** (0.0013)	0.0036** (0.0018)
ω_R	8.8731 (6.1403)	7.5229 (4.9912)	12.093*** (3.3346)	9.4264** (4.3084)
ω_C	49.918 (87.769)	49.843 (118.18)	29.994 (43.68)	2.7464 (4.4)

m	1.6177*** (0.4588)	1.4738*** (0.3441)	0.1569 (0.1245)	1.2451*** (0.2889)
	BM	PP	ST	BC
μ	0.0400 (0.0283)	0.0168 (0.0241)	0.0094 (0.0191)	0.0275 (0.0268)
α	0.0616*** (0.0077)	0.0873*** (0.0083)	0.0721*** (0.0035)	0.0932*** (0.009)
β	0.8781*** (0.0353)	0.8400*** (0.0255)	0.9242*** (0.004)	0.7919*** (0.0323)
θ_R	0.0271*** (0.0053)	0.0277*** (0.0037)	0.019** (0.0092)	0.0291*** (0.003)
θ_C	-0.0015 (0.002)	0.0021 (0.0016)	0.0065*** (0.0019)	-0.0025 (0.0019)
ω_R	15.065** (6.786)	12.45*** (4.531)	5.0028** (2.4862)	18.087*** (3.9919)
ω_C	6.0025 (24.821)	33.704 (66.071)	5.1224*** (1.7876)	1.0023 (2.3009)
m	1.3585*** (0.4241)	0.6755*** (0.194)	0.5485 (0.3537)	1.0946*** (0.2077)

Notes: Robust standard errors are in parentheses; *, **, and *** indicate rejection of the original hypothesis at the 10%, 5%, and 1% levels, respectively.

4.2.3 Analysis of empirical result

The results derived from both the one-factor and two-factor GARCH-MIDAS models indicate that the influence of climate policy uncertainty in China on the indices of non-ferrous metals, building materials, chemicals, paper, and petrochemicals is statistically insignificant. This finding can be interpreted through the lens of industry-specific characteristics, the broader macroeconomic environment in China, and prevailing market expectations. Firstly, the relatively low degree of financialization in certain industries suggests that the performance of their assets in financial markets is primarily driven by physical commodities or traditional industrial applications. Consequently, their commodity prices and market performance exhibit limited sensitivity to market sentiment, investor trading behavior, or fluctuations in macroeconomic policies. Instead, the movement of commodity prices in these sectors is more closely tied to fundamental supply and demand dynamics. For instance, within China's non-ferrous metals sector, while global economic and policy shifts may influence metal asset prices, the speculative component of the market remains relatively subdued. Investor behavior is predominantly guided by fundamental factors,

rather than being driven by financial products or investment-oriented financial speculation(Ren et al,2024^[42]).Secondly, the fundamentals of industries such as building materials and petrochemicals remain stable. China's demand for building materials is closely linked to real estate development and infrastructure construction. Whether in traditional or consumer building materials, only a small portion of industry development is driven by upstream commodities; the majority is propelled by infrastructure projects and the real estate sector. Notably, during years marked by strong policy interventions, such as 2015 and 2020, the government implemented large-scale infrastructure stimulus programs. These robust policy signals have substantially mitigated investors' concerns regarding climate risks.For the petrochemical industry, its relatively low sensitivity to policy changes is attributed to the sustained growth in global energy demand and continued reliance on traditional energy sources. This structural dependence renders policy reversals less costly in an environment of corporate and investor uncertainty. Even in the event of climate policy changes, the underlying demand fundamentals continue to support the stability of the industry(Bouri et al,2022^[43];Uğurlu-Yıldırım et al.2024^[44];Liu et al,2024^[45]).Thirdly, the optimization of industrial structure and technological innovation have contributed to the industry's resilience in the face of potential environmental policy changes. As the world's largest producer of paper and paperboard, China's paper industry has implemented significant measures over the past two decades to reduce environmental pollution, enhance the use of renewable materials and recycling technologies, and gradually transition towards a more sustainable development trajectory in response to technological advancements and shifts in market demand.Yu(2016)^[46] suggests that the productivity of the Chinese paper industry is significantly underestimated when undesirable outputs are not accounted for. As such, stricter environmental laws and regulations can positively influence the industry's ability to internalize climate policy pressures into production processes through enhanced internal management practices.Fourth, China's environmental regulatory policies remain in a developmental phase, characterized by a divergence between soft and hard constraints. This regulatory uncertainty helps to mitigate stock price volatility in high-pollution industries, as the lack of fully enforced, stringent regulations reduces the immediate impact of potential policy shifts.Taking the carbon trading market as an example, China's carbon trading system began as a pilot program in seven local provinces in 2011, with the national carbon emissions trading market only becoming operational in 2021. Although the carbon quota allocation mechanism combines both free allocation and auctioning, free allocation remains dominant in sectors such as chemicals and building materials, where more than 95% of allocations are provided without charge. As a result, the actual financial impact on enterprises is relatively limited, and in the short term, climate policy uncertainty does not significantly influence market volatility in these industries(Song et al,2022^[47];Zhang et al,2019^[48]).

Uncertainty surrounding China's climate policy has had a considerable impact on the transportation, steel, and power sectors, amplifying the volatility of sector indices.Firstly, the industries exhibit a 'lock-in effect' and 'path dependence.' The transportation, steel, and power sectors are traditional industries with high carbon emissions, which have historically relied on conventional energy sources and technologies. The high costs associated with their technological and green transformation make it difficult for these industries to adapt to new climate policies. In the face of climate policy uncertainty, corporate decision-makers are likely to defer investment decisions and technological upgrades. As a result, these industries become more vulnerable to risks, contributing to increased volatility in their stock market

performance(Unruh, 2000[49]).Secondly, investor sentiment plays a crucial role in influencing sectoral indices. Climate policy uncertainty exacerbates investor apprehension, thereby heightening stock market volatility.For the transportation sector, particularly the civil aviation industry, which has high carbon emissions and is more vulnerable to climate risks, there is considerable market focus on the potential for industrial transformation towards low-carbon transportation under the influence of climate policies. Similarly, the power industry, as one of the first major emitting sectors to be incorporated into China's carbon emissions trading market compliance cycle, has garnered significant attention from both Chinese society and the research community.Specifically, the rapid expansion of futures and options markets related to China's transportation and steel industries, coupled with the active engagement of companies in the capital markets and the high degree of financialization within these sectors(Asadi et al,2023^[50]), has heightened investor sensitivity to short-term market price fluctuations, making them more vulnerable to policy changes.Third, the transportation, steel, and power sectors are highly capital-intensive and closely tied to infrastructure development and long-term investments. Uncertainties surrounding climate policies—such as changes in government subsidies for renewable energy, carbon tax policies, and adjustments to emission standards—can significantly impact the total factor productivity of these capital-intensive industries. Such uncertainties may directly influence long-term investment decisions, potentially triggering increased market volatility(Ren et al,2022^[2022]).

4.3 Out-of-sample prediction results

Further, this paper employs the rolling window one-step forward forecasting method to examine whether the single factor and double factor GARCH-MIDAS models incorporating the Chinese Climate Policy Uncertainty Index can improve the forecasting accuracy of the models. In this section, the study intervals(N) are divided into in-sample estimation intervals and out-of-sample prediction intervals, where the length of the out-of-sample prediction intervals is the last 1,000 days corresponding to the actual values of the sample of stock volatility of the eight industries, and 1,000 out-of-sample prediction values are available for each model. The lag order K is still set to 24, consistent with the full sample estimation.

4.3.1 Results based on loss function and DM test

In this paper, we compare and analyze the prediction effects of the baseline model (RV-based GARCH-MIDAS model) and the GARCH-MIDAS-CCPU model replaced by Chinese Climate Policy Uncertainty Index, and the double factor GARCH-MIDAS model, and use the loss function to initially discriminate the model prediction accuracy.Hansen (2005^[52]) stated that in order to avoid problems in terms of robustness, it is recommended to use many different forms of loss functions to discriminate, so in this paper, we use six kinds of loss functions to measure the predictive effect of the model, calculated as follows:

$$MSE = \frac{1}{T} \sum_{t=1}^T (\hat{\sigma}_t^2 - \sigma_t^2)^2 \quad (14)$$

$$MAE = \frac{1}{T} \sum_{t=1}^T |\hat{\sigma}_t^2 - \sigma_t^2| \quad (15)$$

$$MSPE = \frac{1}{T} \sum_{t=1}^T \left(\frac{\sigma_t - \hat{\sigma}_t}{\sigma_t} \right)^2 \quad (16)$$

$$MAPE = \frac{1}{T} \sum_{t=1}^T \left| \frac{\sigma_t - \hat{\sigma}_t}{\sigma_t} \right| \quad (17)$$

$$QLIKE = \frac{1}{T} \sum_{t=1}^T \left(\ln \hat{\sigma}_t^2 + \frac{\sigma_t^2}{\hat{\sigma}_t^2} \right) \quad (18)$$

$$R^2 LOG = \frac{1}{T} \sum_{t=1}^T \left(\ln \frac{\sigma_t^2}{\hat{\sigma}_t^2} \right) \quad (19)$$

where $\hat{\sigma}_t^2$ is the conditional volatility predicted by the model, σ_t^2 is the true volatility of the eight industry indices, and T is the length of the out-of-sample prediction interval. The smaller the value of the loss function, the better the model predicts, and the calculation results are shown in Table 5.

In the nonferrous metals, paper, building material, petrochemical and chemical industries, both the GARCH-MIDAS-CCPU model and the double factor GARCH-MIDAS model have lower prediction effects than the benchmark model, and there is no improvement in prediction accuracy. The prediction effect of GARCH-MIDAS-CCPU model in transportation and steel industries is lower than that of the benchmark model, but the prediction accuracy of the double factor GARCH-MIDAS model is significantly improved compared with that of the benchmark model. The predictive effects of both the GARCH-MIDAS-CCPU model and the double factor GARCH-MIDAS model in the electric power industry are higher than those of the benchmark model, suggesting that Chinese climate policy uncertainty has a high contribution to improving the predictive accuracy of the index in the electric power industry. From the above results, the impacts of Chinese climate policy uncertainty on the eight industries show obvious industry heterogeneity, which is consistent with the previous conclusions.

Table 5. Out-of-sample prediction accuracy based on loss function

	NFM			PA		
	RV	CCPU	RV+CCPU	RV	CCPU	RV+CCPU
MSE	49.5883	50.0872	50.2333	36.9291	38.4656	38.6519
MAE	4.1699	4.4451	4.5039	2.8073	3.3910	3.4417
MSPE	3.5710	2.8009	2.6905	4.7915	2.7808	2.7265
MAPE	1.0976	1.0140	1.0033	1.0655	0.9463	0.9422

QLIKE	2.2861	2.3012	2.3085	1.8313	1.9123	1.9250
R2LOG	7.3834	7.8775	7.9856	7.9429	9.1486	9.2586
EP			TR			
	RV	CCPU	RV+CCPU	RV	CCPU	RV+CCPU
MSE	13.8120	13.7967	12.8136	13.9116	13.9647	13.9051
MAE	1.7739	1.7063	1.6411	1.6685	1.7313	1.6398
MSPE	6.0540	6.0536	5.6402	5.0290	5.0735	4.9416
MAPE	1.0969	1.0885	1.0370	1.0058	0.9721	0.9332
QLIKE	1.3248	1.3137	1.2306	1.3027	1.3255	1.2987
R2LOG	7.8536	7.8493	7.6966	8.3053	8.6427	8.1432
BM			PP			
	RV	CCPU	RV+CCPU	RV	CCPU	RV+CCPU
MSE	31.8715	31.9303	32.5259	21.4422	21.7313	21.4505
MAE	2.9328	2.9910	3.2332	2.4944	2.5984	2.4576
MSPE	3.8685	3.8040	3.0588	4.8875	4.6788	5.2536
MAPE	1.0710	1.0424	0.9813	1.0674	1.0336	1.0970
QLIKE	1.8770	1.8870	1.9158	1.7934	1.8055	1.7967
R2LOG	8.4363	8.6455	9.1784	7.2392	7.5102	7.1199
ST			BC			
	RV	CCPU	RV+CCPU	RV	CCPU	RV+CCPU
MSE	39.7574	41.3095	38.0447	22.6208	22.9228	23.1058
MAE	3.2001	3.8106	3.1680	2.4633	2.5834	2.7233

MSPE	5.2330	3.3098	3.3044	3.9610	4.1314	3.1331
MAPE	1.0984	0.9770	0.9600	1.0292	0.9981	0.9433
QLIKE	1.8997	1.9818	1.6787	1.7439	1.7710	1.7856
R2LOG	7.1584	8.3453	7.0059	7.3584	7.6865	8.0360

According to Qu and Shen(2020)^[53], the mean value of the loss function is susceptible to certain extreme values on the one hand, and on the other hand, it is unable to determine whether the difference in the predictive performance of the models is statistically significant. Therefore, this paper further uses the DM test to compare the prediction accuracy between models, the original hypothesis is "the prediction ability of the benchmark model is not inferior to the comparison model", if the p value is significant to reject the original hypothesis, then it means that the prediction ability of the comparison model is superior to that of the benchmark model, and the results of the calculation are shown in Table 6. The DM statistics of the GARCH-MIDAS-CCPU model and the double factor GARCH-MIDAS model for the electric power industry are both significant at the 1% level, indicating that the two models can effectively improve the accuracy of volatility prediction in the electric power industry, and the double factor GARCH-MIDAS model for the transportation and steel industries also shows better predictive ability, while the other models do not significantly improve the volatility prediction ability compared to the benchmark model. models do not significantly improve the volatility forecasting accuracy.

Table 6. DM test results

	NFM	PA	EP	TR	BM	PP	ST	BC
RV not								
inferior to	-4.110	-1.743	4.074***	1.959*	-0.374	2.017*	2.074*	-2.358
CCPU								
RV not								
inferior to	-4.545	-2.040	4.902***	2.496**	-3.014	-2.276	4.014***	-2.906
RV+CCPU								

Notes: *, **, and *** indicate rejection of the original hypothesis at the 10%, 5%, and 1% levels, respectively.

4.3.2 DoC test results

Based on the methodology of the aforementioned DoC test, this paper reports in Table 7 the value of the DoC ratio, i.e., the number of correct predictions of the direction of volatility movement as a proportion of the total number of predictions. All prediction models are significant at the 1% level, i.e., all GARCH-MIDAS models are statistically significant in predicting the direction of volatility movement of stocks in the eight industries. More importantly, the DoC values of the double factor GARCH-MIDAS models for the nonferrous metals, paper, building material, and chemical industries are smaller than those of the benchmark model, while the DoC values of the double factor GARCH-MIDAS models for the petrochemical, transportation, steel, and electric power industries are larger than those of the benchmark model, demonstrating higher directional prediction accuracy than the benchmark model. In conclusion, the results of the DoC test are basically consistent with the results of the loss function and DM tests.

Table 7. DoC test results

	NFM	PA	EP	TR
RV	<u>0.5785</u> ***	<u>0.5710</u> ***	0.5643***	0.5768***
CCPU	0.5634***	0.5376***	0.5668***	0.5651***
RV+CCPU	0.5609***	0.5326***	<u>0.5701</u> ***	<u>0.5851</u> ***
	BM	PP	ST	BC
RV	<u>0.5701</u> ***	0.5684***	0.5593***	<u>0.5718</u> ***
CCPU	0.5651***	0.5651***	0.5301***	0.5701***
RV+CCPU	0.5568***	<u>0.5743</u> ***	<u>0.5892</u> ***	0.5559***

Notes: Underlined numbers added indicate the maximum value of the DoC proportion, and *** indicates that the PT statistic rejects the original hypothesis at the 1% level.

4.4 Robustness

4.4.1 Results based on different out-of-sample prediction interval lengths

To enhance the robustness of the conclusions, this paper compares the differences in the constructed forecasting models by changing the length of the out-of-sample forecasting intervals, and in this section, the rolling time window technique with an out-of-sample forecasting interval length of 1,200 days is selected to rerun the DM test and the DOC test of the forecasting results of all the models. As shown in

Tables 8 and 9, the results are generally consistent with the conclusions drawn from the out-of-sample forecasts in the previous section.

Table 8. DM test results (predicted length of 1200 days)

	NFM	PA	EP	TR	BM	PP	ST	BC
RV not inferior to	-3.581	-1.757	4.743***	1.044	-0.118	-0.586	1.368	-1.403
CCPU								
RV not inferior to	-4.077	-1.821	5.163***	2.892**	-2.852	-1.522	4.294***	-2.115
RV+CCPU								

Notes: *, **, and *** indicate rejection of the original hypothesis at the 10%, 5%, and 1% levels, respectively.

Table 9. DoC test results (predicted length of 1200 days)

	NFM	PA	EP	TR
RV	<u>0.7037***</u>	<u>0.6895***</u>	0.6786***	0.7062***
CCPU	0.6870***	0.6586***	0.6795***	0.6978***
RV+CCPU	0.6836***	0.6519***	<u>0.6978***</u>	<u>0.7162***</u>
	BM	PP	ST	BC
RV	<u>0.6920***</u>	0.6836***	0.6811***	<u>0.6920***</u>
CCPU	0.6870***	0.6811***	0.6494***	0.6886***
RV+CCPU	0.6761***	<u>0.7020***</u>	<u>0.6994***</u>	0.6828***

Notes: Underlined numbers added indicate the maximum value of the DoC proportion, and *** indicates that the PT statistic rejects the original hypothesis at the 1% level.

4.4.2 Results based on different lags

Engle et al(2013)^[37] indicated that it is better to select macro information with a lag of 3 years to fit the model. Therefore, in this paper, the lag parameter K=36 is used to reconstruct the model for DM test and DOC test of forecasting results, and the results are shown in Table 10 and Table 11. The GARCH-

MIDAS-CCPU model predictions of transportation and steel industries are significantly improved, indicating that the impact of Chinese climate policy uncertainty on the two industries lasts for a longer period of time, and the other results are basically consistent with the conclusions of the previous paper.

Table 10. DM test results (36-month lag)

	NFM	PA	EP	TR	BM	PP	ST	BC
RV not								
inferior to	-3.686	-1.824	3.651***	2.044*	-6.061	-1.014	2.478**	-2.089
CCPU								
RV not								
inferior to	-4.162	-2.551	4.279***	3.477***	-5.594	-2.385	4.179***	-2.711
RV+CCPU								

Notes: *, **, and *** indicate rejection of the original hypothesis at the 10%, 5%, and 1% levels, respectively.

Table 11. DoC test results (36-month lag)

	NFM	PA	EP	TR
RV	<u>0.5801</u> ***	<u>0.5718</u> ***	0.5526***	0.5793***
CCPU	0.5626***	0.5409***	0.5634***	0.5811***
RV+CCPU	0.5551***	0.5376***	<u>0.5943</u> ***	<u>0.5952</u> ***
	BM	PP	ST	BC
RV	<u>0.5659</u> ***	0.5768***	0.5618***	<u>0.5718</u> ***
CCPU	0.5426***	<u>0.5843</u> ***	0.5634***	0.5693***
RV+CCPU	0.5451***	0.5568***	<u>0.5875</u> ***	0.5651***

Notes: Underlined numbers added indicate the maximum value of the DoC proportion, and *** indicates that the PT statistic rejects the original hypothesis at the 1% level.

4.4.3 Results based on different industry index

As the Chinese stock market offers more diversified industry indices, its industry categorization standards are also different. CITIC Securities is one of the strongest securities companies in China, and its

industry indexes are used by more investors. Therefore, in this paper, non-ferrous metal (CI005003.CI), transportation (CI005024.CI), building material (CI005008.CI), petroleum and petrochemical (CI005001.CI), steel (CI005005.CI), and basic chemical industry (CI005006.CI) in the CITIC primary Industry Index are selected to represent the six industries of non-ferrous metal, aviation, building material, petrochemical, steel, and chemical among the eight industries, and paper (CI005122.CI) and power generation and grid (CI005109.CI) in the CITIC secondary industry index are selected to represent the paper and electric power industries, to continue to examine the predictive ability of Chinese climate policy uncertainty on the eight industries. As shown in Tables 12 and 13, the DoC values of the double factor GARCH-MIDAS model for the petrochemical, transportation, steel, and electric power industries are larger than those of the baseline model, which also exhibit higher directional prediction accuracies than those of the baseline model, which is in line with the previous results.

Table 12. DM test results (CITIC industry index)

	NFM	PA	EP	TR	BM	PP	ST	BC
RV not inferior to CCPU								
RV not inferior to	-3.416	-1.046	3.412***	0.804	-0.713	1.967*	1.984*	-2.132
RV+CCPU								
RV not inferior to	-3.991	0.852	4.630***	2.308**	-2.844	-0.578	3.134***	-3.350

Notes: *, **, and *** indicate rejection of the original hypothesis at the 10%, 5%, and 1% levels, respectively.

Table 13. DoC test results (CITIC industry index)

	NFM	PA	EP	TR
RV	<u>0.6903***</u>	<u>0.5684***</u>	0.5760***	0.5876***
CCPU	0.6886***	0.5409***	0.5801***	0.5726***
RV+CCPU	0.6861***	0.5342***	<u>0.5885***</u>	<u>0.5952***</u>
	BM	PP	ST	BC
RV	<u>0.5785***</u>	0.5843***	0.5684***	<u>0.5801***</u>

CCPU	0.5743***	0.5768***	0.5559***	0.5643***
RV+CCPU	0.5568***	<u>0.5960***</u>	<u>0.5959***</u>	0.5484***

Notes: Underlined numbers added indicate the maximum value of the DoC proportion, and *** indicates that the PT statistic rejects the original hypothesis at the 1% level.

Overall, synthesizing the empirical results from Table 8 to Table 13, it can be seen that whether changing the length of the out-of-sample prediction intervals, based on different lags, or based on the test results of different industry indices, the predictive effect of the double factor GARCH-MIDAS model for the transportation and steel industries can be significantly higher than that of the benchmark model, whereas both the GARCH-MIDAS-CCPU model for the electric power industry and the double factor GARCH-MIDAS models for the electric power industry can both predict significantly better than the benchmark model. Therefore, the impacts of Chinese climate policy uncertainty on specific industries are different, and the mixed-frequency volatility GARCH-MIDAS model after considering the Chinese climate policy uncertainty index has significant predictive ability for the volatility of transportation, steel, and electric power industries.

5. Conclusion

In this paper, we use Word2Vec model to obtain two key terms, "climate" and "policy", and calculate the climate policy uncertainty index based on the text data of six major Chinese newspapers from January 1, 2008 to May 31, 2023, and construct single and double factor GARCH-MIDAS models to investigate the impact of climate policy uncertainty on the volatility of eight industries in China. The empirical results show that Chinese climate policy uncertainty has a weak or no significant impact on the volatility of nonferrous metals, paper, building material, petrochemical, and chemical industries, and has a significant impact on the volatility of transportation, steel, and electric power industries. Compared with the RV-based benchmark model, the models incorporating the Chinese Climate Policy Uncertainty Index (CCPU) for paper and petrochemical industries show only a slight advantage in some of the results; the double factor GARCH-MIDAS models incorporating the CCPU for transportation and steel industries show better predictive ability; and the double factor GARCH-MIDAS-CCPU model or double factor GARCH-MIDAS-CCPU model for the electric power industry have better predictive ability than the RV-based benchmark model. Finally, the robustness test results based on the three dimensions of this paper further support the above conclusions. Compared with existing studies, the set of key terms obtained in this paper based on the deep learning model Word2Vec is able to avoid the problem of subjective selection bias of

keywords to a certain extent, and at the same time, we investigate the impact of Chinese climate policy uncertainty on the volatility of the stock market from the perspectives of different industries, and further find out the heterogeneity of the impact on the volatility of different industries. For industries significantly affected by Chinese climate policy uncertainty, the Chinese Climate Policy Uncertainty Index has a significant predictive ability on their volatility, and this result helps government departments and investors to more accurately grasp the dynamics of volatility, which is of great significance to the optimization of risk management and investment strategies.

Based on the results of the research, this paper makes the following recommendations:

First, decision-making bodies must carefully consider the shifts in public expectations arising from climate policy uncertainty. As China actively advances its "dual-carbon" goals, the introduction and adjustment of climate policies are inevitable. Consequently, the uncertainty associated with these policy changes warrants thorough attention. The implications of climate policy uncertainty can significantly influence investor sentiment, potentially fostering irrational behaviors such as herd behavior and speculative trading. This, in turn, may exacerbate market volatility and facilitate the transmission of instability across markets, ultimately undermining overall market stability. On one hand, it is crucial to continually improve the transparency of climate policy adjustments by establishing a platform for the dynamic and real-time release of policy updates. This platform should facilitate regular interpretations of policy objectives and implementation details, thereby mitigating short-term market fluctuations driven by misinterpretations. Additionally, a "policy buffer period" mechanism should be implemented, providing a public consultation phase prior to significant policy changes, allowing time for stakeholders to adjust and reducing uncertainty. Special attention should be given to mitigating panic caused by major climate events or extreme crises, with efforts focused on stabilizing public expectations in the short term. On the other hand, it is essential to adjust the intensity and direction of climate policies in a timely manner, taking into account the specific characteristics of each industry and sector. This involves introducing differentiated countermeasures tailored to distinct timeframes and formulating a phased emission reduction roadmap. Clear and transparent rules should be established, such as carbon quotas and subsidies for technological transformation, to reduce decision-making uncertainty for enterprises. The formulation of climate policies should be scientifically grounded, considering not only the effectiveness of emission reductions but also the uncertainty stemming from potential policy changes. Maintaining the continuity and stability of climate policies is critical to minimizing market risks associated with frequent policy adjustments and avoiding long-term adverse impacts.

Second, investors should adopt a dynamic approach to constructing their investment portfolios, adjusting their investment decisions promptly in response to both climate policy uncertainty and prevailing market conditions. During periods of heightened climate policy uncertainty, substantial capital flows may occur between sectors driven by differing investment objectives, such as risk aversion and profit-seeking, thereby exacerbating market volatility. This necessitates a proactive strategy, with investors carefully managing their exposure to sectors most vulnerable to policy shifts and aligning their portfolios with evolving market dynamics. Different industries, and even individual companies within those industries, are impacted by climate policy uncertainty in varying degrees. Investors must closely monitor climate policy uncertainty and assess its potential positive and negative effects on their investment portfolios from a fundamental perspective. Risk-averse investors should seek to avoid assets that could be subject to increased volatility, thereby minimizing unnecessary risks. At the same time, they can strategically position their portfolios to capitalize on favorable policy shifts, such as subsidies for clean energy industries, by constructing active and well-informed portfolios that align with the country's green and low-carbon transition. This approach not only reduces risk but also offers opportunities for investing in the nation's sustainable development.

Finally, in response to the challenges posed by climate uncertainty, enterprises should, on one hand, strengthen internal climate risk management by implementing "climate scenario analysis" frameworks. This would allow them to assess the potential impacts of different policy scenarios on cash flow and capital expenditures. At the same time, companies should optimize their capital structure by leveraging green bonds or ESG financing mechanisms, which can help mitigate financing costs in the face of policy uncertainty. This strategic approach not only ensures financial resilience but also positions enterprises to benefit from the growing demand for sustainable investments, enhancing their long-term viability in an evolving policy landscape. Enterprises should actively foster technological innovation and drive green transformation by increasing investment in the research and development of low-carbon technologies, green energy solutions, energy conservation, and environmental protection initiatives. Furthermore, they should promote cross-industry collaboration, enabling high-emission industries to jointly develop low-carbon solutions in partnership with clean technology companies and share the benefits of technological spillovers. Externally, enterprises should enhance information disclosure and transparency by regularly communicating their strategies and measures for adapting to changes in climate policy to the public and investors. Transparent information disclosure not only helps mitigate market panic triggered by uncertainty but also strengthens investor confidence, facilitating a more stable and predictable investment

environment.

References

- [1] Paroussos L, Mandel A, Fragkiadakis K, et al. Climate clubs and the macro-economic benefits of international cooperation on climate policy[J]. *Nature Climate Change*,2019,9(3):542-546.
- [2] Tian H, Long S B, Li Z X. Asymmetric effects of climate policy uncertainty, infectious diseases-related uncertainty, crude oil volatility, and geopolitical risks on green bond prices[J]. *Finance Research Letters*,2022,48.
- [3] Chen Z, Zhang L, Weng C. Does climate policy uncertainty affect Chinese stock market volatility?[J]. *International Review of Economics & Finance*, 2023, 84: 369-381.
- [4] Friedman M. The Role of Monetary Policy[J]. *The American Economic Review*,1968,58(1).
- [5] Rodrik D. Policy uncertainty and private investment in developing countries[J]. *Journal of Development Economics*,1991,36(2):229-242.
- [6] Hassett K A, Metcalf G E. Investment with Uncertain Tax Policy: Does Random Tax Policy Discourage Investment?[J]. *The Economic Journal*, 1999, 109(457).
- [7] Pástor L, Veronesi P. Political uncertainty and risk premia[J]. *Journal of Financial Economics*,2013,110(3):520-545.
- [8] Born B, Pfeifer J. Policy risk and the business cycle[J]. *Journal of Monetary Economics*,2014,68.
- [9] Hassan T A, Stephan H, Laurence V L, et al. Firm-Level Political Risk: Measurement and Effects[J]. *The Quarterly Journal of Economics*,2019,134(4):2135-2202.
- [10] Wesseh P K, Benjamin N I, Lin B Q. The coordination of pumped hydro storage, electric vehicles, and climate policy in imperfect electricity markets: Insights from China[J]. *Renewable and Sustainable Energy Reviews*,2022,160.
- [11] Borenstein S, Bushnell J, Wolak F A, et al. Expecting the Unexpected: Emissions Uncertainty and Environmental Market Design[J]. *American Economic Review*, 2019,109(11):3953-3977.
- [12] Liu Z H, Ding Z H, Duan Z P. Does climate policy uncertainty exacerbate extreme risk spillovers across energy markets?[J]. *Systems Engineering-Theory and Practice*, 2023, 43(06):1651-1667.
- [13] Ding S J, Liang Z P, Chen J Y. Research on the Spillover Effects among Climate Policy Uncertainty, Energy, and Metal Markets from the Time-Frequency Perspective[J]. *Systems Engineering-Theory and Practice*, 2023, 43(08):2222--2235.
- [14] Khanh H. How does corporate R&D investment respond to climate policy uncertainty? Evidence from heavy emitter firms in the United States[J]. *Corporate Social Responsibility and Environmental Management*,2022,29(4):936-949.

- [15] Ren X H, Qin J N, Dong K Y. How Does Climate Policy Uncertainty Affect Excessive Corporate Debt? The Case of China[J]. *Journal of Environmental Assessment Policy and Management*, 2022, 24(02).
- [16] Kunreuther H, Heal G, Allen M, et al. Risk management and climate change[J]. *Nature Climate Change*, 2013, 3(5).
- [17] Batten S. Climate change and the macro-economy: A critical review[J]. *Social Science Electronic Publishing*, 2018.
- [18] Ren X H, Li J Y, He F, et al. Impact of climate policy uncertainty on traditional energy and green markets: Evidence from time-varying granger tests[J]. *Renewable and Sustainable Energy Reviews*, 2023, 173: 113058.
- [19] Guo J Q, Long S B, Luo W J. Nonlinear effects of climate policy uncertainty and financial speculation on the global prices of oil and gas[J]. *International Review of Financial Analysis*, 2022, 83.
- [20] Chen X, Chen X, Xu L H, et al. Attention to climate change and downside risk: Evidence from China[J]. *Risk Analysis*, 2023, 43(5), 1011-1031.
- [21] Xu Y A, Li M, Yan W, et al. Predictability of the renewable energy market returns: The informational gains from the climate policy uncertainty[J]. *Resources Policy*, 2022, 79.
- [22] He M X, Zhang Y J. Climate policy uncertainty and the stock return predictability of the oil industry[J]. *Journal of International Financial Markets, Institutions & Money*, 2022, 81.
- [23] Yan W L, Cheung A. The dynamic spillover effects of climate policy uncertainty and coal price on carbon price: Evidence from China[J]. *Finance Research Letters*, 2023, 53, 103400.
- [24] Coeure B. Monetary policy and climate change[Z]. Speech at scaling up green finance, Deutsche Bundesbank, Berlin, 2018, Nov. 8th.
- [25] Xu X, Huang S P, Lucey B M, et al. The impacts of climate policy uncertainty on stock markets: Comparison between China and the US[J]. *International Review of Financial Analysis*, 2023, 88.
- [26] Liang C, Muhammad U, Ma F, et al. Climate policy uncertainty and world renewable energy index volatility forecasting[J]. *Technological Forecasting & Social Change*, 2022, 182.
- [27] Chen Z L, Zhang L, Weng C. Does climate policy uncertainty affect Chinese stock market volatility? [J]. *International Review of Economics and Finance*, 2023, 84.
- [28] Baumöhl E, Lyócsa Š. Directional predictability from stock market sector indices to gold: A cross-quantilogram analysis [J]. *Finance Research Letters*, 2017, 23 152-164.
- [29] Lu X J, Ma F, Wang J Q, et al. Singlehanded or joint race? Stock market volatility prediction [J]. *International Review of Economics and Finance*, 2022, 80 734-754.

- [30] Blyth W, Bradley R, Bunn D, et al. Investment risks under uncertain climate change policy [J]. *Energy Policy*, 2007, 35 (11): 5766-5773.
- [31] Benhabib J, Liu X, Wang P. Sentiments, financial markets, and macroeconomic fluctuations[J]. *Journal of Financial Economics*, 2016, 120(2): 420-443
- [32] Hirshleifer D, Teoh S H. Herd behaviour and cascading in capital markets: A review and synthesis[J]. *European Financial Management*, 2003, 9(1): 25-66.
- [33] Baker S R, Bloom N, Davis S J. Measuring Economic Policy Uncertainty[J]. *The Quarterly Journal of Economics*, 2016, 131(4): 1593-1636.
- [34] Huang Y, Luk P. Measuring economic policy uncertainty in China[J]. *China Economic Review*, 2020, 59(C).
- [35] Gavrilidis K. Measuring climate policy uncertainty[J]. SSRN Working Paper, 2021. Available at SSRN: <http://dx.doi.org/10.2139/ssrn.3847388>
- [36] Mikolov T, Chen K, Corrado G, et al. Efficient estimation of word representations in vector space[J]. preprint arXiv:1301.3781, 2013.
- [37] Engle R F, Ghysels E, Sohn B. Stock market volatility and macroeconomic fundamentals[J]. *Review of Economics and Statistics*, 2013, 95(3): 776-797.
- [38] Diebold F, Mariano S. Comparing Predictive Accuracy[J]. *Journal of Business & Economic Statistics*, 1995, 13(3).
- [39] Degiannakis S, Filis G. Forecasting oil price realized volatility using information channels from other asset classes[J]. *Journal of International Money and Finance*, 2017, 76.
- [40] Pesaran M H, Timmermann A. A Simple Nonparametric Test of Predictive Performance[J]. *Journal of Business & Economic Statistics*, 1992, 10(4).
- [41] Bai J C, Huang Y F, Gao Z A, et al. Research on the Impact of Economic Policy Uncertainty and Cryptocurrency Market volatility: Empirical Analysis Based on GARCH-MIDAS Model[J]. *Operations Research and Management Science*, 2022, 31(05): 183-189.
- [42] Ren X, Fu C, Tao L, et al. Climate Policy Uncertainty and Financialization of Non-Ferrous Metals in China[J]. Available at SSRN 4907000.
- [43] Bouri E, Iqbal N, Klein T. Climate policy uncertainty and the price dynamics of green and brown energy stocks[J]. *Finance Research Letters*, 2022, 47: 102740
- [44] Uğurlu-Yıldırım E, Dinç-Cavlak Ö. Climate policy uncertainty, media coverage of climate change, and energy markets: New evidence from time-varying causality analysis[J]. *Energy and Climate Change*,

2024, 5: 100134.

[45] Liu F, Su C W, Tao R, et al. Does economic and climate policy uncertainty matter the oil market?[J]. Resources Policy, 2024, 95: 105188.

[46] Yu C, Shi L, Wang Y, et al. The eco-efficiency of pulp and paper industry in China: an assessment based on slacks-based measure and Malmquist–Luenberger index[J]. Journal of cleaner Production, 2016, 127: 511-521.

[47] Song X, Wang D, Zhang X, et al. A comparison of the operation of China's carbon trading market and energy market and their spillover effects[J]. Renewable and Sustainable Energy Reviews, 2022, 168: 112864.

[48] Zhang F, Fang H, Song W. Carbon market maturity analysis with an integrated multi-criteria decision making method: A case study of EU and China[J]. Journal of Cleaner Production, 2019, 241: 118296.

[49] Unruh G C. Understanding carbon lock-in[J]. Energy policy, 2000, 28(12): 817-830.

[50] Asadi M, Tiwari A K, Gholami S, et al. Understanding interconnections among steel, coal, iron ore, and financial assets in the US and China using an advanced methodology[J]. International Review of Financial Analysis, 2023, 89: 102789.

[51] Ren X, Zhang X, Yan C, et al. Climate policy uncertainty and firm-level total factor productivity: Evidence from China[J]. Energy Economics, 2022, 113: 106209.

[52] Hansen P R. A Test for Superior Predictive Ability[J]. Journal of Business & Economic Statistics, 2005, 23(4): 365-380.

[53] Qu S, Shen W. The impact of investor attention on stock market volatility based on the LSTHAR model[J]. Chinese Journal of Management Science, 2020, 28(07): 23-34.